F16AG427-VS
VARIABLE SPEED DRIVE

11,000V
RD WH BU
FCF 100A
QAC 100A 100A 400A
BCT1 50 / 1A 2.5VA 5P20
BCT2 50 / 1A 2.5VA 5P20
BCT3 50 / 1A 2.5VA 5P20
BCN 50 / 1A 1.5VA 10P10
BAD
PFV1 RD RD RD
QCE 400A
EARTH BAR
X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13
2000-EL003-F16-10016 SHEET 14

SYSTEM INTERCONNECTION DIAGRAM 2000-EL005-B02-10001
L1 L2 L3
LINE SIDE INCOMING SUPPLY SHEET 6
OUTGOING MOTOR SUPPLY SHEET 6
T1 T2 T3
CABLE SCREEN EARTHED
M 3~
F16-AG-427
ROUGHER FLOTATION CELL 4 AGITATOR
3,300V
500 kW

F16-AG-427
MOTOR HEATER TERMINAL BOX
F16-HT-547
X2 54 53
F16HT547-L
F16HT547-N
0.3kW
ANTI-CONDENSATION HEATER
MOTOR SPACE HEATER CIRCUIT SHEET 6

X81 (PART OF)
RTD TEMPERATURE MONITOR RELAY
SHEET 6

F16-AG-427
MOTOR RTD TERMINAL BOX
X81
PT100 RTD INPUT CH1: 13 COM 14 - 15
PT100 RTD INPUT CH2: 16 COM 17 - 18
PT100 RTD INPUT CH3: 19 COM 20 - 21
PT100 RTD INPUT CH4: 22 COM 23 - 24
PT100 RTD INPUT CH5: 25 COM 26 - 27
PT100 RTD INPUT CH6: 28 COM 29 - 30
PT100 RTD INPUT CH7: 31 COM 32 - 33
PT100 RTD INPUT CH8: 34 COM 35 - 36
F16AG427A-RD
F16AG427A-WH
F16AG427A-BK
F16AG427B-RD
F16AG427B-WH
F16AG427B-BK
F16AG427C-RD
F16AG427C-WH
F16AG427C-BK
F16AG427D-RD
F16AG427D-WH
F16AG427D-BK
F16AG427E-RD
F16AG427E-WH
F16AG427E-BK
RTD
MOTOR WINDING RTD (RD PHASE) (TD1)
MOTOR WINDING RTD (WH PHASE) (TD3)
MOTOR WINDING RTD (BU PHASE) (TD5)
MOTOR BEARING RTD (DRIVE END) (BD10)
MOTOR BEARING RTD (NON DRIVE END) (BD20)

F16-SW-011 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10016 SHEET 7 3A & 3B
X3 3 1139 X2 67
X3 17 4203 X2 68
K41 K41
MV CONTACTOR ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10016 SHEET 10 4A
X2 4 1179 X2 63
X2 25 1199 X2 64
K41 K41 K41
MV INPUT ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD TRIP 2000-EL003-F16-10016 SHEET 10 3A
X2 3 1219 X2 65
X2 24 1239 X2 66
K41 K41 K41
MV INPUT TRIP CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10016 SHEET 8 5A & 5D
X2 56 1259 X2 70
X2 44 1279 X2 69
A30 (PART OF)
X2 15 TRIP X2 16
ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F16-DB-101
2000-F16-TLD-EL-10054
RD WH BU BK
R17 25A
W17
B17
EARTH BAR
415VAC
F16DB101-R17
F16DB101-W17
F16DB101-B17
F16DB101-BK17
X4 1 2 3 4 PE1
VSD AUXILIARY POWER SUPPLY SHEET 6

F16-DB-201
2000-F16-TLD-EL-10057
RD BK
R3 10A
EARTH BAR
240VAC
F16DB201-R3
F16DB201-BK3
X4 5 6 PE2
VSD UPS POWER SUPPLY SHEET 6

TO FLOTATION 11kV SWITCHROOM
F16-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F16-ETH-001
2000-F16-TER-IC-10002
CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6
X47
ETHERNET ADAPTOR RJ45
A2 (PART OF)
DI-2B 9

F16AG427-LCS
LOCAL CONTROL STATION
LOS
START
X86 1 2 3 4
X1 1 2 3 4 5 6 E
F16AG427VS-1409
F16AG427VS-1419
F16AG427VS-1429
F16AG427VS-1439
F16AG427VS-1459
F16AG427VS-1469
SAFE TORQUE OFF CIRCUIT SHEET 6
X2 13 14
24VDC CONTROL SHEET 6

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F16AG427-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.

LEGEND

- 11kV SWITCHBOARD TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL

PANEL A08

2000-F16-SCM-EL-10115_Rev1

F16-SW-011 F16-AG-427 scavenger flot. cell 4 agi VSD feeder schematic diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCOM GROWTH PROJECT\03 DELIVERABLES\EL\F16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SCM-EL-10115.DWG 09SEP25 6:39AM HOTSEAT04